

River geomorphology and hydrologic climate as predictors of elusive summer-run coho salmon in southeast Alaska

Alaskan climate patterns are projected to be both significantly warmer (average increase of 8.1°F – 14.2°F) and generally wetter by the end of the 21st century, with precipitation patterns expected to be characterized by increased variability and more precipitation falling as rain rather than snow (U.S. Department of Agriculture 2025). These shifts are anticipated to have substantial impacts on hydrologic processes throughout the region, which will in turn affect the survival of freshwater species and populations within their native ranges (Mizukami et al. 2022, Leppi et al. 2014). One such population, summer-run coho salmon (*Oncorhynchus kisutch*), are an elusive group that has heretofore been largely overlooked in both academic and management contexts. In the face of shifting hydrology, it is essential to better understand where these fish are found and where they are most threatened by changes to their habitat.

The study and management of rare and elusive species or populations often comes at excessive cost, which in turn may result in poor understanding of these groups' life histories and roles in larger ecosystems (Thompson 2013). These costs can be reduced given some foreknowledge of the likelihood associated with encountering populations at specific locations. Research focused on predicting occupancy based on habitat features is, therefore, not uncommon (Vale et al. 2016, Burnham et al. 2021, Zlonis et al. 2022). Throughout their native range, coho are typically associated with a spawning season beginning in the fall and concluding late in the year (Theriault et al. 2010, Quinn 2018, Alaska Department of Fish and Game 2025). The late-season run timing of fall coho tends to coincide with heavier rainfall and higher water levels in streams, providing an advantage to coho attempting to reach spawning grounds inaccessible to other Pacific salmon species (Quinn 2018). However, throughout this same range, there are known instances of far less abundant summer coho runs, most notably in the Sol Duc and Quillayute rivers on Washington's Olympic Peninsula (National Park Service 2025). While these summer-run coho runs are known to exist, research specifically focused on them is paltry. Boolean searches of academic literature in *Web of Science* for the terms "summer coho", "summer-run coho", and "summer run" AND "coho", return very few results, with most concerned either with juvenile coho rearing during summer, or with other salmonid species (such as steelhead and/or Chinook) being discussed in the same work. Summer coho runs also often go unmonitored by fisheries managers, with efforts typically concentrated later in the season when the vast majority of spawning coho return. Due to the costs associated with monitoring these elusive populations in remote areas, managers and biologists would be greatly aided by foreknowledge regarding where and when they are likely to be found. (T. Fish and A. Piston – Alaska Department of Fish and Game, August 22, 2024).

Anecdotal evidence suggests that southeast Alaska's summer coho are smaller than their fall-run cousins and often found high in stream systems above treacherous falls and cascades. Their smaller size, potentially the result of less time spent at sea feeding

during summer months, may enable these populations to navigate in-stream obstructions which act as barriers to other fish, therefore accessing spawning habitat that fall-run coho are unable to reach. Any partial barriers must therefore be substantial enough that larger fall-run coho are unable to traverse them, despite the benefit of higher flows during fall months (T. Fish and A. Piston – Alaska Department of Fish and Game, August 22, 2024). It is also possible that stream temperature may play a significant role in habitat suitability, as spawning grounds at higher elevation would typically see cooler temperatures earlier in the season, thereby necessitating earlier runs (B. Romey – Romey Riverscape Science and D. Martin – Martin Environmental LLC, June 30, 2025). The identification of partial barriers below with suitable coho spawning habitat, along with temperature regimes associated with summer coho habitat, could together provide a strong indication of where survey efforts might best be directed to assess the extent of these populations. Further, identifying locations where suitable temperatures are likely to be impacted most by a warming climate could shed light on where these populations are most vulnerable.

This research focuses on identifying summer-run coho spawning habitat in southeast Alaska, using predictive statistical models to estimate the likelihood of these populations inhabiting stream systems. Southeast Alaska is characterized by steep mountains descending rapidly to glacially carved fjords, leading to streams with drastic elevation changes occurring over relatively short distances, providing ample opportunities for occurrences of the partial barriers described above (Smith 2016). My predictive models incorporate presence-absence data of known partial barriers with detailed stream geomorphology data. Data providing spatial locations of known partial barriers and summer-coho populations are provided by the Alaska Department of fish and Game. Stream-level geomorphology data come from TerrainWorks, a private research team dedicated to the development of “virtual watersheds” using remote sensed digital elevation models. These data include estimates of stream flow, gradient, and valley constraint (the ratio of valley-floor width to maximum stream width), which have been shown to be robust proxies for coho intrinsic habitat potential (Burnett et al. 2007). Finally, stream temperature data come from the Southeast Alaska Watershed Coalition, including both historical records and projections of temperatures under various future climate regimes. Binomial generalized linear models (GLMs) have been shown to perform well when predicting the spatial distribution of a species based on presence-absence data, with the extent of the data available influencing this model’s predictive ability (Carlos-Junior et al. 2020).

This work will fill knowledge gaps in both the ecological understanding of summer-run coho as well as regulatory agencies’ ability to monitor and manage these fish. Funding from the PCC Graubard Fellowship will enable the development of models able to accurately predict which physical features and temperature regimes are most likely to indicate the presence of summer-run coho, making these populations easier to study and enabling fisheries managers to regulate commercial and sport fishing activities to ensure the long-term survival of southeast Alaska’s summer-run coho.

References

- Alaska Department of Fish and Game. 2025. Prince of Wales Management Area – Fishing Information. Available at: <https://www.adfg.alaska.gov/index.cfm?adfg=ByAreaSoutheastPrinceofWales.fishinfo> (accessed May 10, 2025)
- Burnett, K. M., G. H. Reeves, D. J. Miller, S. Clarke, K. Vance-Borland, K. Christiansen. 2007. Distribution of salmon-habitat potential relative to landscape characteristics and implications for conservation. *Ecological Applications* 17(1): 66-80.
- Burnham, R. E., D. A. Duffus, T. Ross. 2021. Remote sensing and mapping habitat features pertinent to fin whale life histories in coastal and offshore waters of Vancouver Island, British Columbia. *Journal of Experimental Marine Biology and Ecology* 537, 11 pp.
- Carlos-Junior, L. A., J. C. Creed, R. Marrs, R. J. Lewis, T. P. Moulton, R. Feijo-Lima, M. Spencer. 2020. Generalized Linear Models outperform commonly used canonical analysis in estimating spatial structure of presence/absence data. *PeerJ* 8: 21 pp.
- Leppi, J. C., D. J. Rinella, R. R. Wilson, W. M. Loya. 2014. Linking climate change projections for an Alaskan watershed to future coho salmon production. *Global Change Biology* 20: 1808-1820.
- Mizukami, N., A. J. Newman, J. S. Littell, T. W. Giambelluca, A. W. Wood, E. D. Gutmann, J. J. Hamman, D. R. Gergel, B. Nijssen, M. P. Clark, J. R. Arnold. 2022. New projections of 21st century climate and hydrology for Alaska and Hawai'i. *Climate Services* 27: 15 pp.
- National Park Service. 2025. Olympic National Park, Washington – Coho Salmon. Available at: <https://www.nps.gov/olym/learn/nature/coho-salmon.htm> (accessed May 12, 2025).
- Quinn, T. P. 2018. *The Behavior and Ecology of Pacific Salmon and Trout, Second Edition*. University of Washington Press, 547 pp.
- Smith, M. A. 2016. *Ecological Atlas of Southeast Alaska*. Audubon Alaska, 228 pp.
- Theriault, V., G. R. Moyer, M. A. Banks. 2010. Survival and life history characteristics among wild and hatchery coho salmon returns: how do unfed fry differ from smolt releases? *Canadian Journal of Fisheries and Aquatic Sciences* 67(3): 486-497.
- Thompson, W. 2013 *Sampling Rare or Elusive Species – Concepts, Designs, and Techniques for Estimating Population Parameters*. Island Press, 429 pp.
- Vale, C. G., J. C. Campos, T. L. Silva, D. V. Goncalves, A. S. Sow, F. Martinez-Freiria, Z. Boratynski, J. C. Brito. 2016. *Biogeography and conservation of mammals from the West Sahara-Sahel: an application of ecological niche-based models and GIS*. *Hystrix-Italian Journal of Mammalogy* 27(1), 10 pp.

U.S. Department of Agriculture. 2025. Climate Hubs – Alaska and a Changing Climate. Available at: <https://www.climatehubs.usda.gov/hubs/northwest/topic/alaska-and-changing-climate> (accessed October 2, 2025).

Zlonis, E. J., R. Deo, J. B. Berdeen. 2022. LiDAR and multispectral imagery predict the occurrence of tree cavities suitable for a cavity-nesting duck. *Remote Sensing in Ecology and Conservation* 8(2), 191-207.